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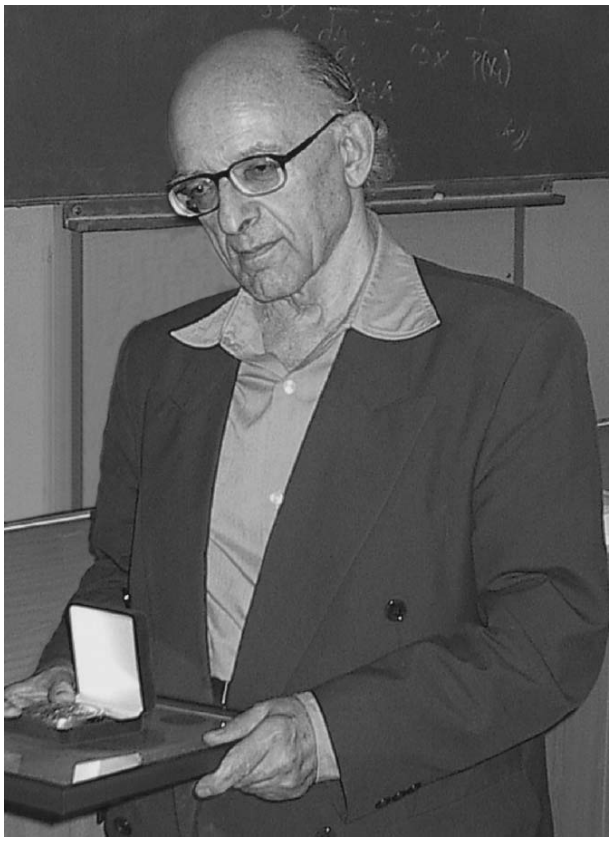

I.M. Sobol receiving the commemorative medal "For a Contribution to Approximate Integration Theory," in Krasnoyarsk in 2003.

On the 80th Birthday of Il'ya Meierovich Sobol

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August 15, 2006, marked the 80th birthday of Professor Il'ya Meierovich Sobol, a principal researcher at the Institute of Mathematical Modeling of the Russian Academy of Sciences. Sobol is an outstanding Russian mathematician internationally renowned for his fundamental works in applied mathematics and mathematical modeling. The editorial board of this journal congratulates one of our long-time active authors and experts in computational mathematics. It is sufficient to remember that the first issue of this journal, which was released in 1961, included two of Sobol's papers. Over the years, this journal has published 23 of his papers (sometimes coauthored) on various problems in computational mathematics. He is a world-famous expert in Monte Carlo methods and their generalizations. Many problems concerning quadrature formulas, statistical trials, experiment design, sensitivity analysis, etc., have been solved using these techniques and their more advanced versions proposed by Sobol. Some of his books are popular with many research prac-

titioners. Some books were translated into other languages and became well known abroad. Despite his age, Sobol continues active research, which is illustrated by several of his works published in 2006.

Sobol was born on August 15, 1926, in Panevezhas (Lithuania). At the beginning of the Great Patriotic War, his family was evacuated to Izhevsk, where Sobol graduated with distinction from high school in 1944. In the same year, he entered the Faculty of Mechanics and Mathematics of Moscow State University and graduated from it with distinction in 1948. In his student years, Sobol was actively engaged in solving various mathematical problems. His first scientific works concerning ordinary differential equations were published in renowned mathematical journals in 1948. Some of his subsequent studies were also devoted to this subject. In 1949, Sobol joined Laboratory No. 8 of the Geophysical Complex Expedition at the Institute of Geophysics of the USSR Academy of Sciences, which was established by a decree of the USSR government and later merged with the Institute of Applied Mathematics of the USSR Academy of Sciences. At this institute, Sobol worked for many years, taking part in the computations for the first Soviet atomic and hydrogen bombs, for which he was awarded the Medal for Labor Valor and the Order of the Honor Badge.

In 1954, Sobol focused on Monte Carlo methods, and most of his numerous subsequent works were concerned directly or indirectly with this area. Some of his studies beyond this subject are also well known and frequently referred to in the scientific literature. An example is his joint work with Academician A.A. Samarskii on the computation of temperature waves, where the blowup regime describing heat localization was found for the first time. In 1958, Sobol published his paper "Pseudorandom Numbers for the Strela Computer," where he constructed the first (in our country) original pseudorandom number generator. The possibility of a deterministic approach to the calculation of expectations was discussed in that paper. Later, this approach was called the quasi-Monte Carlo method. Sobol coauthored a well-known book on Monte Carlo methods issued in 1962, which has been translated into several languages and still retains its importance. In connection with the problem of pseudorandom numbers, Sobol focused on the theory of uniform distributions of such numbers. He was the first to use the Haar functions in mathematical applications. In this area, he wrote his doctoral dissertation "The Method of Haar Series in the Theory of Quadrature Formulas," which he defended in 1972 at the Institute of Mathematics (Siberian Division, USSR Academy of Sciences) in Novosibirsk. This issue was also addressed in his well-known monograph Multidimensional Quadrature Formulas and Haar Functions, which came out in 1969. Sobol gave much attention to the theory and applications of Monte Carlo methods in various scientific branches (for example, in astrophysics). In 1973, he published his monograph Numerical Monte Carlo Methods, where he introduced the new concepts of constructive dimension, vector statistical weights, etc. Especially popular is the new class of uniformly distributed LN_T-sequences, which are now known as Sobol sequences. He showed that they are superior over other sequences in terms of many characteristics. These sequences are widely used in the evaluation of integrals and in optimization, experiment design, etc. Sobol showed that Monte Carlo algorithms can be considerably accelerated by using terms of such sequences instead of random numbers. Such methods are now known as quasi-Monte Carlo methods. In a series of joint studies with scientists from the Institute of Engineering Science of the USSR Academy of Sciences, Sobol developed an efficient method for solving multicriterial optimization problems, which was widely used to choose machine design parameters.

Especially worth noting is Sobol's contribution to the solution of various astrophysical problems, where he successfully applied Monte Carlo methods, etc. For nearly ten years, he participated in mathematical simulation of X-ray and gamma spectra of compact relativistic objects. He solved various problems concerning the transmission of neutrons, photons, protons, etc. The methods developed by Sobol together with his collaborators have been successfully applied in our country and abroad.

A series of Sobol's works focused on the error analysis of computational algorithms for anisotropic classes of functions of several variables. He was the first to study optimal algorithms in anisotropic classes of functions and to show that many "optimal" algorithms are ineffective in practice and should be replaced by so-called uniformly optimal algorithms.

In sensitivity theory, Sobol introduced global sensitivity indices, which are intended to analyze the structure of multivariate functions defined by a computing code rather than analytically. This concept has been used in various science areas for many years. Sobol coauthored a book on sensitivity theory published abroad in 2000.

In 1968, Sobol published the popular-science book Monte Carlo Method, which went through several editions and was translated into five languages. A revised version of this book came out in the USA in 1994.

Sobol has published about 170 scientific papers, monographs, and textbooks. Many of his scientific papers were issued in renowned scientific journals in our country and abroad. His works have a very high citation index.

For many years, Sobol worked as a professor at the Department of Mathematical Physics of the Moscow Engineering Physics Institute, where he lectured and supervised undergraduate and graduate studies.

Sobol serves on the editorial board of an international journal on Monte Carlo methods and their applications. He participated in many scientific conferences on Monte Carlo methods, optimization techniques, and sensitivity theory. He is a member of the New York Academy of Sciences.

Especially worth noting are Sobol's activities in the Journal of Computational Mathematics and Mathematical Physics. His reviews of submitted manuscripts are especially appreciated by the editorial board. He is a recognized expert on applications of Monte Carlo methods.

The editorial board of this journal, its staff members, and authors wish Il'ya Meierovich Sobol good health, many happy years ahead, new scientific achievements, and success in all his activities.

LIST OF SOBOL'S BASIC PUBLICATIONS

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